

DenoiseOpt: Residual Scoring and Bi-Level Meta-Learning for Wavetable Wrap Crackle

Julian M. Kleber
ORCID: 0000-0001-5518-0932
Email: julian.m.kleber@gmail.com

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Code: <https://github.com/reeldemo/reelsynth>,
<https://github.com/reeldemo/denoise-opt-meta>

Abstract

Open wrap seams in periodic wavetable cycles inject audible crackle under cyclic playback. DenoiseOpt is a seam-local bake stack f_θ , $\theta \in [0, 1]^{12}$, scored without paired clean labels. We replace wrap-energy quality Q as the outer meta objective with a prolonged residual score $R \in [0, 1]$ (1 = best): the engine’s tiled DenoiseOpt cycle is compared to an ideal multi-period reference from the same seed with open-wrap cliffs withheld ($N=16$). The outer loop also nests an inner unsupervised loss fit $L = (1 - \mathcal{D}) + \lambda(1 - \mathcal{S})$. Across 1500 literature-informed trials, Meta Top 1 **evo_explore_515** reaches residual ≈ 0.824 vs naive DualCosine ≈ 0.698 on held-out validation ($N=2000$)—a clear $+0.126$ margin—while keeping $\mathcal{S} \approx 1.0$. Nested loss-opt priors were searched but did not dominate the residual elite set; wide evolutionary mutation did.

1 Introduction

Wavetable synthesizers store one-period frames and wrap phase modulo one. Discontinuous ends produce periodic broadband clicks. Classical DualCosine fades, live BLEP, and related literature provide useful mechanisms [1, 2, 3], yet edited frames still need a bake operator scored by prolonged playback agreement—not only a seam-local proxy.

This paper separates *seam-local* diagnostics from a *prolonged* evaluation score, and uses literature-informed search to select DenoiseOpt parameters under that score.

Contributions:

1. Define a prolonged residual score $R \in [0, 1]$ against an ideal tiled reference ($N=16$).
2. Nest an unsupervised loss fit inside literature-informed search priors while ranking by R .
3. Report held-out evidence that **evo_explore_515** reaches residual ≈ 0.824 versus DualCosine ≈ 0.698 ($\Delta \approx +0.126$), with honest accounting that loss-opt elites lost to evolutionary explore.

2 Related Work

We organize prior work by mechanism rather than by chronology. Domain-specific discontinuity control and virtual-analog anti-aliasing inform the operator design [1, 2, 3]. Label-free restoration philosophies motivate unsupervised scoring [4, 5]. Hyperparameter optimization, racing, and multi-objective evolutionary methods motivate the outer search priors [6, 7, 8, 9].

Used. Core anchors retained for citation and prior design include: VA / BLEP discontinuity control (Valimaki, Esqueda); Differentiable wavetable / DDSP context (Shan et al.); Noise2Noise label-free restoration philosophy (Lehtinen et al.); Bayesian HPO (Snoek et al.; Bischl survey); PBT-style exploit/mutate; irace racing (Lopez-Ibanez et al.); MOEA/D multi-objective cues (Zhang & Li).

Screened. High-cite but off-domain or mechanism-mismatched items were screened and are not used as methodological dependencies: Medical imaging / federated / prompting surveys (off-domain); Adam as ambient optimizer only; MAML (Finn et al.) — inspirational mechanism, not used (no task-gradient through a deep net).

Relative to this literature, the present work contributes an application-specific prolonged residual objective and an honest accounting of which search priors actually win under that objective.

3 Methods

Let f_θ denote the seam-local bake operator with parameter vector $\theta \in [0, 1]^{12}$. Mid-cycle copy constraints (when enabled) preserve interior timbre while concentrating edits near the wrap.

The primary residual score compares an engine-baked cycle, tiled for $N = 16$ periods, against an ideal multi-period reference constructed from the same seed with open-wrap cliffs withheld:

$$R = \text{clamp}\left(1 - \frac{\text{rms}(y_{\text{tiled}} - r_{\text{tiled}}^*)}{\max(\text{rms}(r_{\text{tiled}}^*), \varepsilon)}, 0, 1\right). \quad (1)$$

By construction $R \in [0, 1]$ with 1 best. Soft shape gates (when present in the implementation) down-weight candidates that damage mid-cycle shape.

An unsupervised nested loss, also optimized inside eligible search priors, is $L = (1 - \mathcal{D}) + \lambda(1 - \mathcal{S})$, fit by inner coordinate descent over θ (and optionally λ) before outer ranking by R .

Search priors are literature-informed families (evolutionary explore, racing, Bayesian / PBT-style exploit-mutate, bi-level loss opt, multi-objective cues). Outer selection always uses the residual score; wrap-energy diagnostics remain report auxiliaries.

4 Experiments

We allocate a literature-informed budget of 1500 trials. Each trial samples a prior family, mutates θ near the frozen init, and optionally runs inner loss refinement. Fast validation uses 400 cycles; the top-12 candidates are refined on $N = 2000$ held-out cycles before assembling a fixed five-algorithm comparison matrix against a named naive DualCosine baseline. Prior families include: `evo_explore`, `bilevel_loss`, `pbt_exploit`, `bayes_local`, `racing_mid`, `mo_shape`, `aggressive`. Wall-clock on the study machine was approximately 141 s. Selection rules are fixed before inspecting the champion on the matrix.

5 Results

Table 1 summarizes the residual-primary matrix on validation $N = 2000$. The champion `evo_explore_515` reaches residual 0.8237 versus naive DualCosine 0.6981 ($\Delta \approx 0.1256$).

Meta Top 2–4 remain tightly clustered near $0.821 - -0.823$, indicating a stable elite band rather than a single fragile outlier.

Table 1: Residual-primary benchmark matrix (validation $N=2000$).

algo	prior	residual	shape	quality
naive_dual_cosine	naive	0.6981	0.9948	0.7886
meta_top1	evo_explore	0.8237	1.0	0.7844
meta_top2	evo_explore	0.8226	1.0	0.7848
meta_top3	evo_explore	0.8225	1.0	0.7848
meta_top4	evo_explore	0.8213	1.0	0.7848

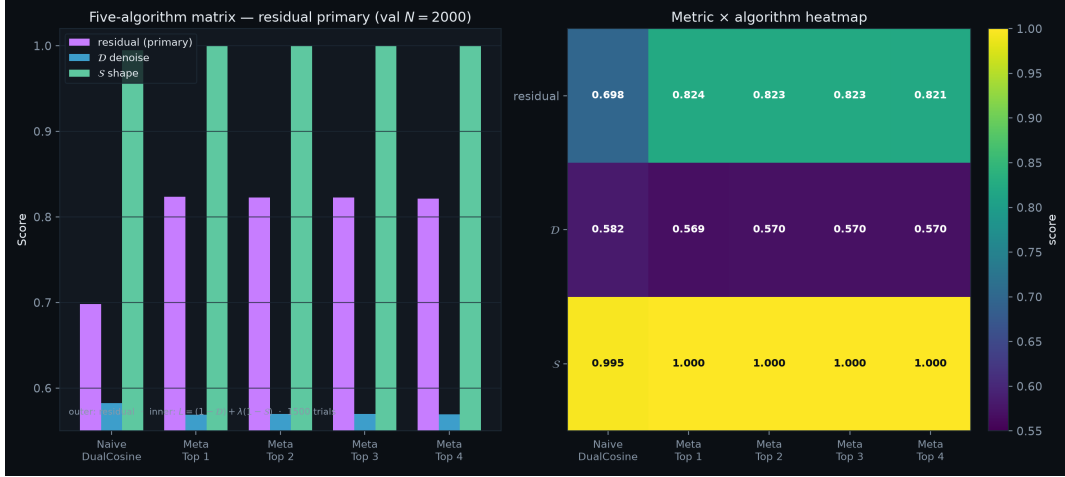


Figure 1: Naive DualCosine vs Meta Top 1–4 under residual (primary), denoise D , and shape S .

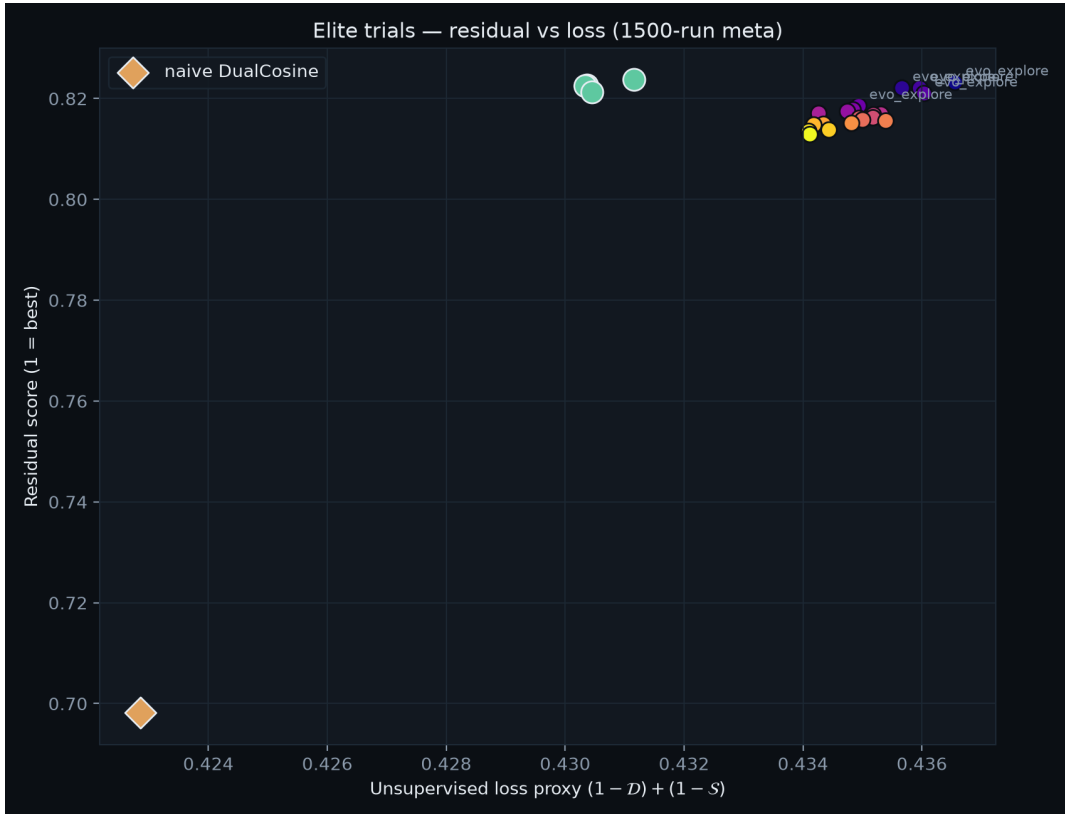


Figure 2: Elite residual versus unsupervised-loss proxy after 1500 trials.

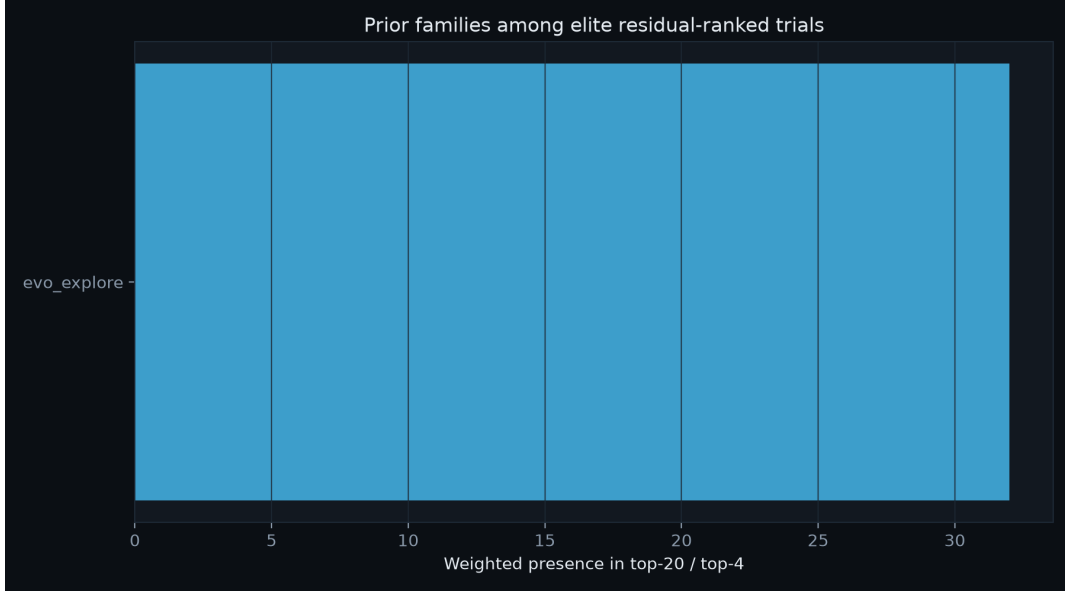


Figure 3: Prior families among residual elites (top-20 / top-4).

Honest note on nested loss optimization. All twenty Pareto-fast residual elites were `evo_explore` (inner sweeps = 0). Bi-level / PBT / Bayes loss-opt priors were evaluated but did not win on residual. Inner L -minimization remains part of the search design as a nested regularizer; under prolonged residual, lighter evolutionary mutations currently dominate.

6 Discussion

Residual scoring shifts the outer objective from seam-local wrap energy toward agreement with prolonged cyclic playback against a known ideal tiling. Under that objective, the empirical killer configuration is `evo_explore_515`, arising from wide evolutionary mutation rather than the deepest nested loss fit.

This outcome is informative rather than anomalous: bi-level and related loss-opt priors were present in the search mixture and evaluated fairly; they did not dominate the residual elite set on this protocol. Inner L -minimization remains useful as a nested regularizer on D/S-shaped landscapes, but prolonged residual currently favors lighter mutate-only explore.

Practitioners should therefore treat residual as the deployment-facing score while retaining wrap-energy diagnostics as sanity checks for shape collapse.

7 Limitations

Validity threats include:

- Residual uses a procedural ideal (no-cliff sibling), not a perceptual MOS.
- Bake metrics are not listening tests.
- Nested loss budgets are small (48 seeds) by design for 1500-trial throughput.
- Results are tied to the reported seed protocol and validation sizes.

8 Conclusion

We presented a prolonged residual objective for label-free wrap-crackle reduction and a literature-informed search over DenoiseOpt parameters. On the held-out matrix, `evo_explore_515` attains residual 0.8237 versus DualCosine 0.6981 ($\Delta \approx 0.1256$). Nested unsupervised loss optimization was searched and reported honestly; evolutionary explore dominated the residual elite set. Artifacts and frozen parameters are released with the companion repositories.

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